

One Interval, Two Cutoffs

AP Statistics · Topic 7.3 (CED 4.3) · B: 2027-01-15 · E: 2027-03-03

[STAR] TODAY'S PROBLEM

Suppose a library wants the mean setup time for all 800 kiosk transactions in a month below 45 seconds and hopes to advertise a mean below 42 seconds. An SRS of 25 has the shown summary. A 95% one-sample t -interval is (39.923, 44.877) seconds.

Mina: “I would compare each cutoff with the full interval and keep the conclusion about the population mean.”

Gabe: “Because the interval contains times below both cutoffs, I think it supports both claims; I also read 95% as the share of transaction times inside it.”

Kiosk sample

Summary	Value
Population	800
n	25
Mean (s)	42.4
SD (s)	6.0
Shape	Symmetric; no outliers

FIRST TAKE — before the video (5 min)

Which approach is more defensible? Cite an endpoint or identify the quantity being estimated.

[BOOK] LET THE ENTIRE INTERVAL GOVERN THE CLAIM (keep on the page)

A $C\%$ confidence interval estimates a population mean, not the spread of individual observations. We are $C\%$ confident that the interval captures the population mean represented by the random sample; the fixed interval either contains that mean or it does not. A value inside the interval remains plausible. To support $\mu < c$, the entire interval must lie below c ; an endpoint merely falling below c is not enough. With other features held fixed, a larger sample tends to narrow a mean interval, while a higher confidence level widens it.

Finish after the video:

DOK 1 (a) Define the population mean and locate 42 and 45 relative to the interval.

DOK 2 (b) Verify random, 10 percent, and shape conditions. Reproduce the standard error, margin of error, and interval; interpret it in context.

DOK 3 (c) Adjudicate the approaches. Use both endpoints to evaluate the two less-than claims, explain why the cutoff locations matter, distinguish the population mean from individual times, and trace what the rejected interpretation could lead the library to advertise.

Frame: Relative to the interval, 42 is _____ and 45 is _____; together, those locations support _____.

Frame: The interval estimates _____ rather than _____, so the advertisement should _____.

EXIT — turn this in

One thing the video changed about my first take: _____